Bachelor of Computer Applications (Hons. Game Development)
Semester- III

Subject: Database Management System

Assignment 3: Database Design Analysis

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1. Introduction

This analysis continues the Library Management System designed in Assignments 1 and 2. The system uses 11 related tables to manage members, staff, publishers, authors, categories, book titles, physical copies, loans, reservations, and fines. The purpose of separating these entities is to reduce repeated facts and keep the data consistent.

2. Data redundancy and anomalies

Data redundancy occurs when the same fact is stored repeatedly. For example, if a single table stored a book title together with every copy, loan, member, publisher, and author, the publisher name and book title would be repeated for each physical copy and every borrowing transaction. The submitted design avoids much of this repetition by storing those facts in separate tables and connecting them through keys.

3. Functional dependencies

A functional dependency X → Y means that a value of attribute set X determines exactly one value of attribute set Y. The following dependencies are supported by the proposed schema. Each primary key determines the non-key attributes in its own table.

Relation	Main functional dependencies
Members	member_id → name, email, phone, address, membership_date, membership_status; email → member_id (UNIQUE)
Staff	staff_id → name, email, phone, designation, date_joined; email → staff_id (UNIQUE)
Publishers	publisher_id → name, phone, address
Authors	author_id → name, nationality, date_of_birth
Categories	category_id → category_name, description; category_name → category_id (UNIQUE)
Books	book_id → title, isbn, publisher_id, category_id, publication_year, language; isbn → book_id (UNIQUE)
BookCopies	copy_id → book_id, rack_number, status
Book_Author	(book_id, author_id) → no additional non-key attributes; pair uniquely identifies the link
Loans	loan_id → copy_id, member_id, staff_id, issue_date, due_date, return_date

Reservations	reservation_id → book_id, member_id, reservation_date, status
Fines	fine_id → loan_id, amount, reason, paid_status, payment_details; loan_id → fine_id (UNIQUE)

4. Normalization analysis

First Normal Form (1NF): Attributes store single values rather than repeating lists. Multiple authors are represented as separate rows in Book_Author, not as a comma-separated list in Books.

Second Normal Form (2NF): The only composite primary key is (book_id, author_id) in Book_Author. The table contains no non-key attributes, so no non-key attribute depends on only part of the composite key.

Third Normal Form (3NF): Publisher, category, author, member, staff, and book details are stored in their own relations. Non-key details are not copied into Loans or BookCopies. This reduces transitive dependencies such as loan_id → copy_id → book_id → publisher_id.

5. Design improvements and integrity considerations

- Keep the current separation of master, junction, and transaction tables.
- Enforce UNIQUE on member/staff email, ISBN, category name, and the fine's loan_id where these values are intended to identify unique records.
- Use CHECK constraints for status values, non-negative fine amounts, and valid loan dates.
- Add indexes to frequently searched foreign keys and fields such as Loans.member_id, Loans.copy_id, Books. title and Reservations. status when performance warrants them.
- The current BookCopies.status and Loans.return_date represent related availability information.
- Add application logic or a trigger/transaction to prevent a copy from having two simultaneous open loans and keep copy status synchronized with loan activity.
- In Assignment 1, the relationship Loans → Fines is described as one-to-one. The schema implements this by making Fines. loan_id UNIQUE. If the library later needs separate fine items for late return and damage on one loan, this rule would need to change.

6. Conclusion

The 11-table design reduces redundancy by storing each entity's facts in one place and using foreign keys for relationships. The remaining risks are mainly

business-rule consistency across tables, which should be handled with constraints and transactional logic.